

Lab Manual

IT 141-Fundamentals of Computer Programming

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TURBO C : IDE

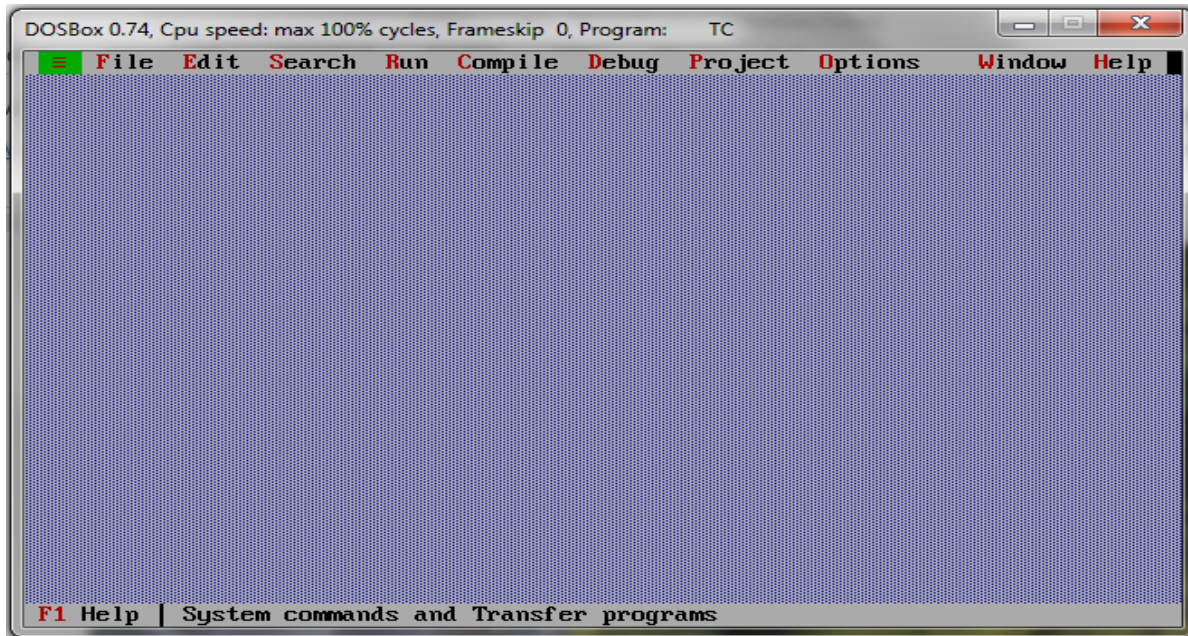
➤ Introduction

The C Developing Environment is a screen display with windows and pull-down menus. the program listing, error messages and other information are displayed in separate windows. The menus may be used to invoke all the operations necessary to develop the program, including editing, compiling, linking, and debugging and program execution.

➤ How to start Turbo C?

Double click on the icon of Turbo C.



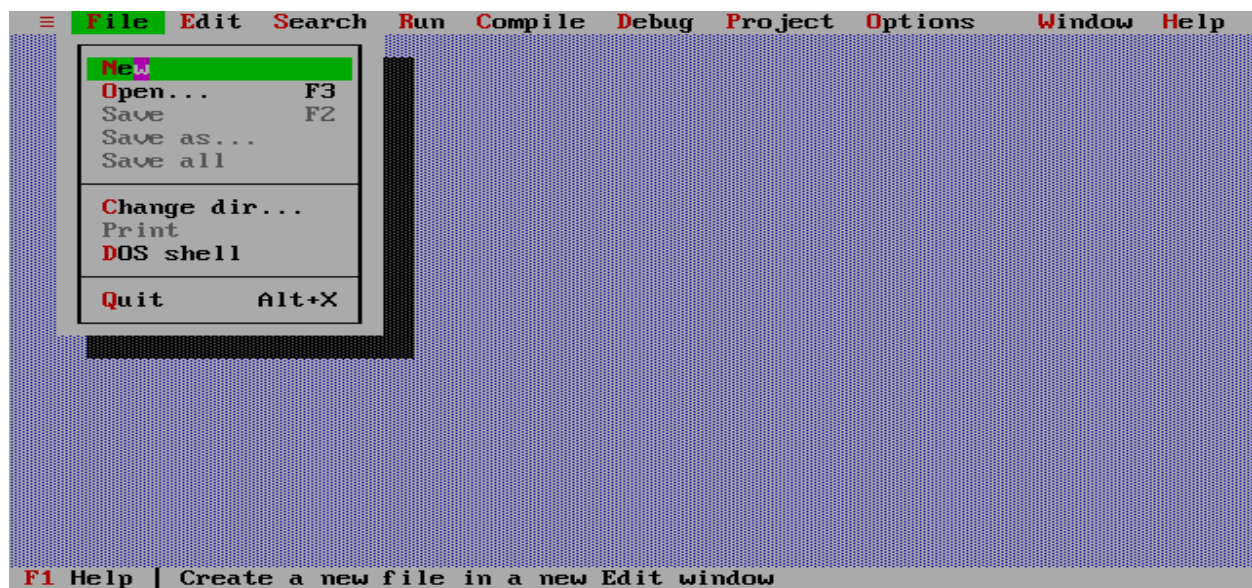


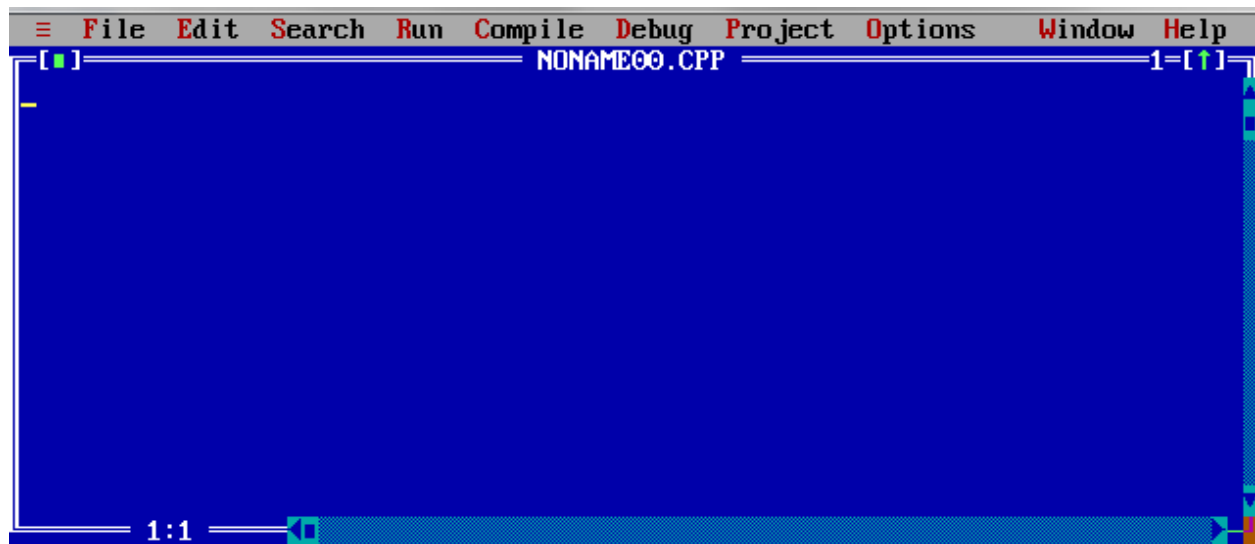
➤ Invoking the Turbo C

The default directory of Turbo C compiler is **C:\TC\BIN**.

➤ Opening New Window

To type a program, we need to open file menu and click “new”.





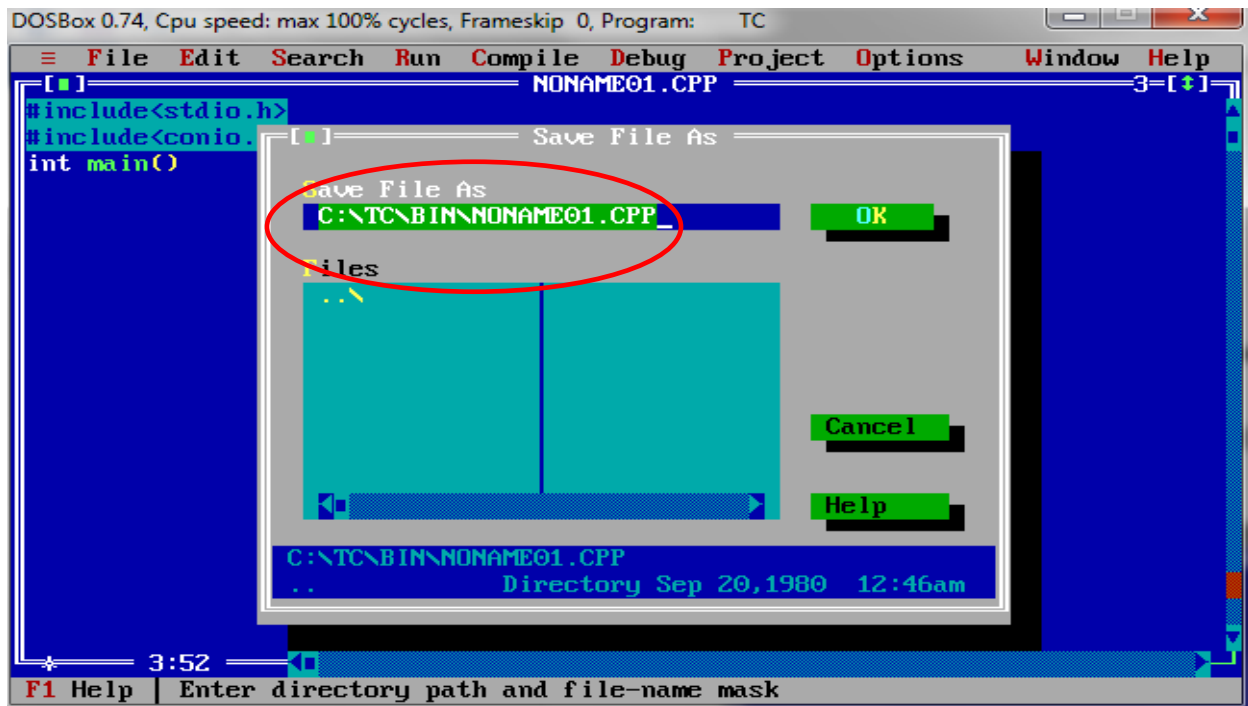
ALT+ENTER – Open Turbo C in full screen mode.

➤ Writing a Program

When the Edit window is active, the program may be typed.

➤ Saving a Program

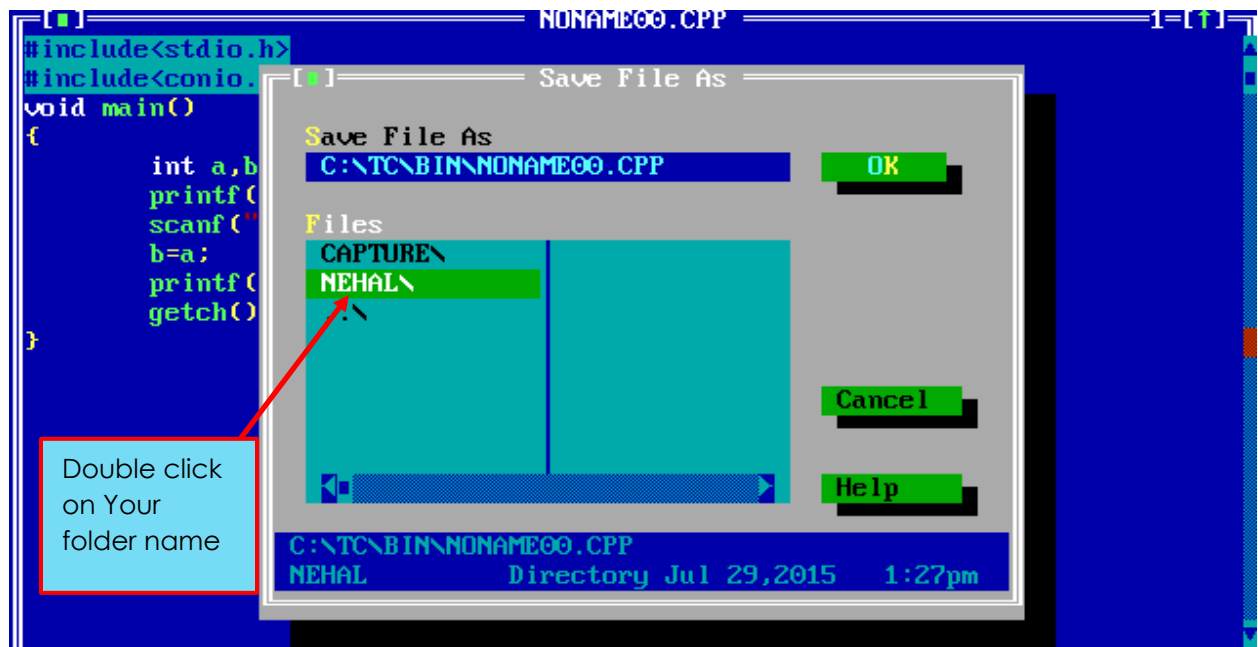
By default program name is “NONAME__.CPP”. For save program, go to file menu select **SAVE** command or pressing **F2 key**. Save your program with unique file name & .C extension. **Ex: “hello world . c”**

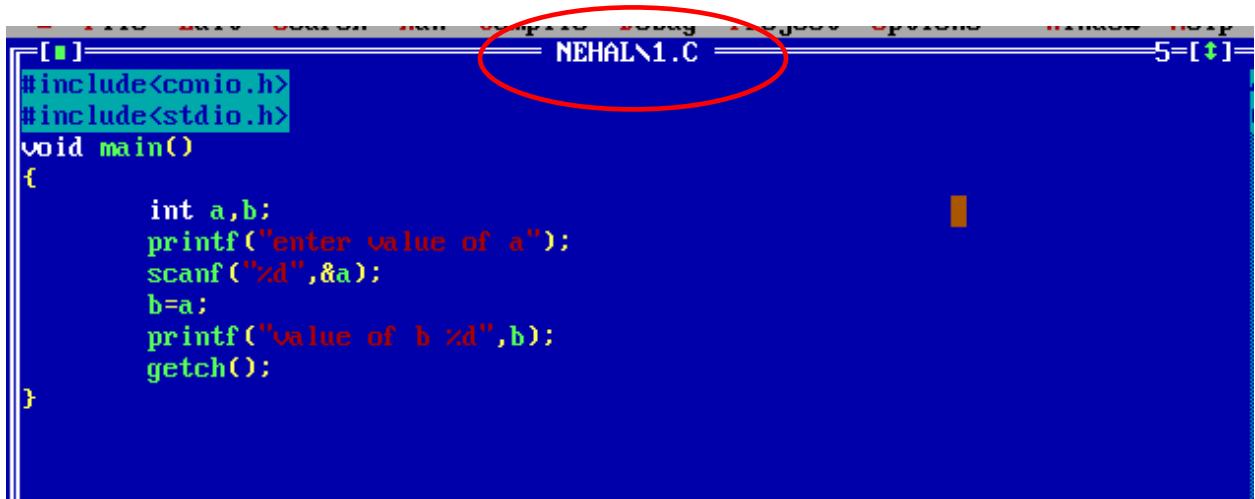
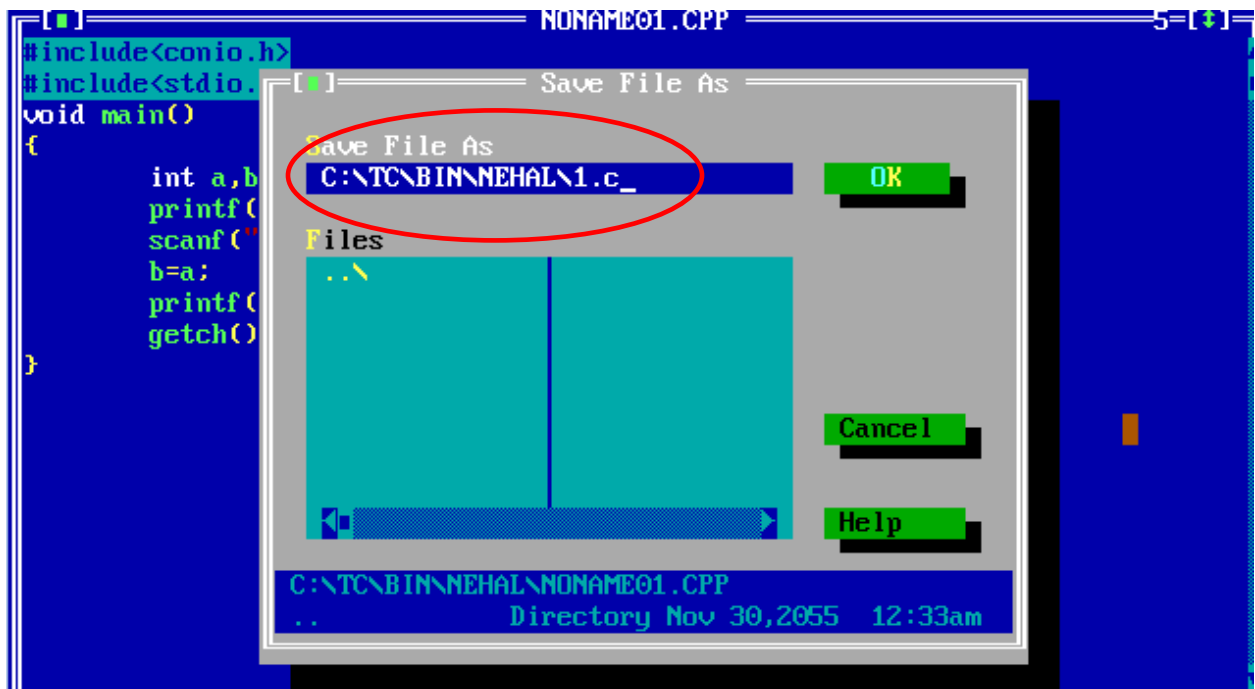


➤ Make your own folder

Step 1: Go to C Drive in which Turbo C installed. Double click on TC Folder.

Step 2: Go to BIN folder and make new folder with your ID No. (EX: 15IT001)





➤ Making an Executable File

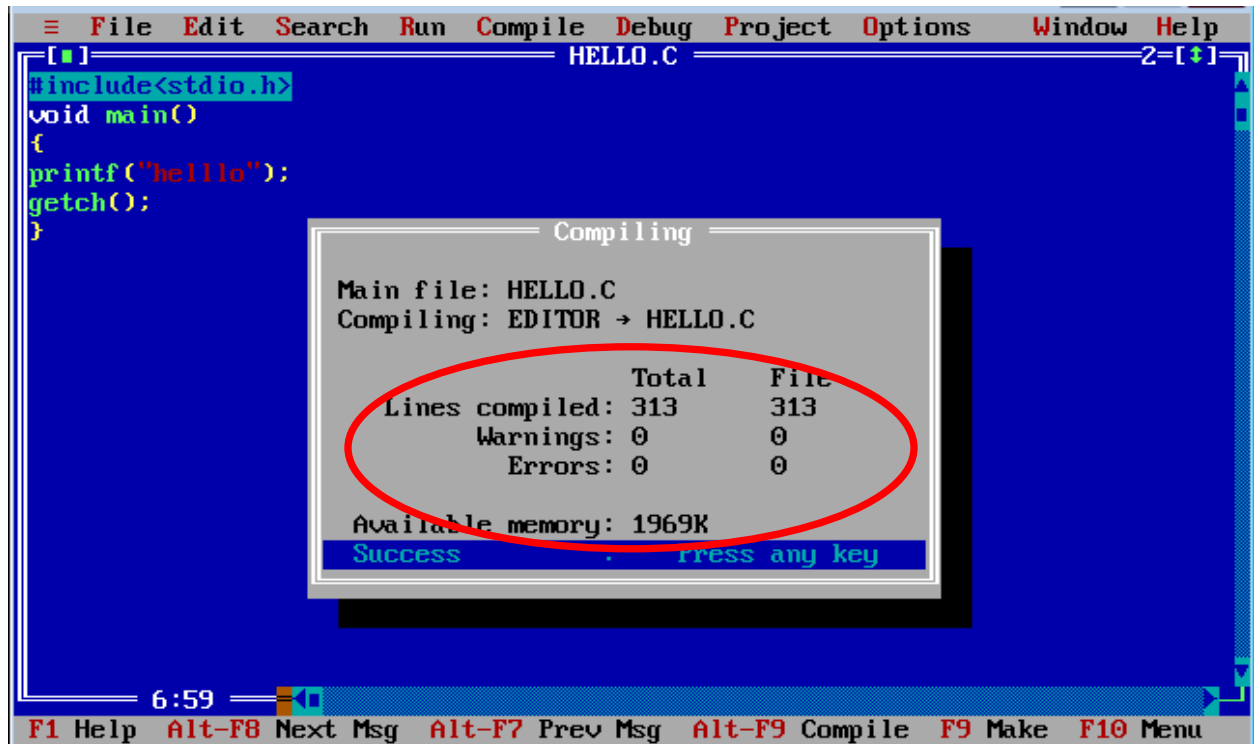
The source file is required to be turned into an executable file. This is called “Making” of the **.exe** file. The steps required to create an executable file are:

1. Create a source file, with a **.c extension**.
2. Compile the source code into a file with the **.obj extension**.
3. **Link your . obj file** with any needed libraries to produce an executable program.

All the above steps can be done by using *Run option* from the menu bar or using key combination **Ctrl+F9** (By this linking & compiling is done in one step).

➤ Compiling and Linking

In the Turbo C, compiling and linking can be performed together in one step. There are two ways to do this: you can select **Make EXE** from the **compile menu**, or you can press the **[F9]** key. For compilation, press **[Alt + F9]**. One box is open with Lines compiled, Warnings and Errors.



```
#include<stdio.h>
void main()
{
printf("hello");
getch();
}
```

Compiling

Main file: HELLO.C
Compiling: EDITOR → HELLO.C

	Total	File
Lines compiled:	313	313
Warnings:	0	0
Errors:	0	0

Available memory: 1969K
Success . . . press any key

6:59

F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

```
File Edit Search Run Compile Debug Project Options Window Help
HELLO.C
#include<stdio.h>
main()
{
printf("hello")
}
```

Compiling

Main file: HELLO.C
Compiling: EDITOR → HELLO.C

	Total	File
Lines compiled:	312	312
Warnings:	1	1
Errors:	2	2

Available memory: 1969K
Errors :

4:17

F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

```
File Edit Search Run Compile Debug Project Options Window Help
HELLO.C
#include<stdio.h>
main()
{
printf("hello")
}
```

Message

Compiling HELLO.C:
•Error HELLO.C 5: Statement missing ;
Error HELLO.C 5: Compound statement missing }
Warning HELLO.C 5: Function should return a value

➤ Correcting Errors

If the compiler recognizes some error, it will let you know through the Compiler window. You will see that the number of errors are listed. The compiler will not only tell you what you did wrong, they'll point you to the exact place in your code where you made the mistake.

➤ Executing a Programs

If the program is compiled and linked without errors, the program is executed by selecting **Run** from the Run Menu or by pressing the **[Ctrl+F9]** key combination.

➤ Exit Turbo C

To exit from the IDE, select **Exit** from the File Menu or press **Alt+X** Combination.

Practical List : 1

1. Write a program that prints your mailing address in the following form:

First line: Name

Second line: Address (No, Street/Society Name)

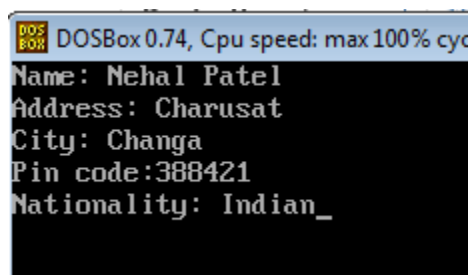
Third line: City, Pin code

Fourth line: Nationality

// Use of Printf() and \n

```
#include<stdio.h>
void main()
{
    clrscr();
    printf("Name: Nehal Patel:\n");
    printf("Address: Charusat:\n");
    printf("City: Changa\n");
    printf("Pin code:388421:\n");
    printf("Nationality: Indian");
    getch();
}
```

Output:



```
DOS
BOX DOSBox 0.74, Cpu speed: max 100% cyc
Name: Nehal Patel
Address: Charusat
City: Changa
Pin code:388421
Nationality: Indian_
```

2. Write a program to calculate Simple Interest

$$SI = (P * R * N)/100;$$

Where, P = principle amount, R = Rate of interest, N = Period

// simple interest

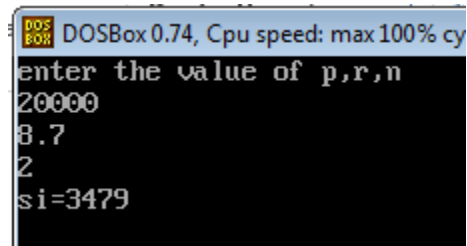
```
#include<stdio.h>
#include<conio.h>
```

```

void main()
{
    int si,p,n;
    float r;
    clrscr();
    printf("enter the value of p,r,n");
    scanf("%d %f %d",&p,&r,&n);
    si=p*r*n/100;
    printf("si=%d",si);
    getch();
}

```

Output:



```

DOSBox 0.74, Cpu speed: max 100% cy
enter the value of p,r,n
20000
8.7
2
si=3479

```

3. Write a program to compute a volume of a cylinder. Use symbolic constant.

Formula: $V = \pi r^2 h$

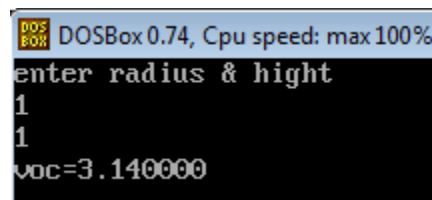
// volume of a cylinder

```

#include<stdio.h>
#include<conio.h>
#include<math.h>
#define pi 3.14
int main()
{
    float r,h,voc;
    clrscr();
    printf("enter radius & hight");
    scanf("%f %f",&r, &h);
    voc=pi*r*r*h;
    printf("voc=%4f",voc);
    getch();
    return 0;
}

```

Output:



```

DOSBox 0.74, Cpu speed: max 100%
enter radius & hight
1
1
voc=3.140000

```

4. Write a program to read the price of an item in decimal form(like 15.95) and print the output in paise(like 1595 paise)

```
#include<conio.h>
#include<stdio.h>
void main()
{
    float rs;
    int PAISA;
    clrscr();
    printf("Enter the Rs. you want to convert to paisa :- ");
    scanf("%f",&rs);

    PAISA=rs*100;
    printf("\n The PAISA of you Rs. equal to :- %d",PAISA);

    getch();
}
```

Output:

Enter the Rs. you want to convert into paisa :- 12.50

The paisa of Rs. is equal to :- 1250

5. Write a program to calculate Net Salary. User has to input Basic Salary and Output should be:

Enter Basic Salary: 5000 (e.g. 5000)

Allowances:

DA = 70% of Basic Salary

HRA = 7% of Basic Salary

MA = 2% of Basic Salary

TA = 4% of Basic Salary

Deduction:

PF = 12% of Basic Salary

IT = any value (e.g. 500)

Net Salary = Basic Salary + Allowances – Deduction

```
#include<stdio.h>
#include<conio.h>
#define p printf
int main()
{
    float bs=5000,da,hra,ma,ta,pf,it=500,ns,allow,deduction;
    clrscr();
    da = 0.7*bs;
    p("\n da is: %f",da);
```

```

        hra = 0.07*bs;
        p("\n hra is: %f",da);
        ma = 0.02*bs;
        p("\n ma is: %f",da);
        ta = 0.04*bs;
        p("\n ta is: %f",da);
        allow = da+hra+ma+ta;
        p("\n allow is: %f", allow);
        pf = 0.12*bs;
        p("\n pf is: %f",da);
        deduction = pf-it;
        p("\n deduction is : %f ", deduction);
        ns = bs +allow - deduction;
        p("\n Net salary is: %f ", ns);
    getch();
    return 0;
}

```

Output:

...

```

da is: 3500.000000
hra is: 3500.000000
ma is: 3500.000000
ta is: 3500.000000
allow is: 4150.000000
pf is: 3500.000000
deduction is : 100.000000
Net salary is: 9050.000000

```

7. Write a program to print the ASCII value of a given character and vice-versa.

```

#include <stdio.h>
int main()
{
    char c;
    printf("Enter a character: ");
    // Reads character input from the user
    scanf("%c", &c);
    // %d displays the integer value of a character
    // %c displays the actual character
    printf("ASCII value of %c = %d", c, c);
    return 0;
}

```

Output:

Enter a character: G
 ASCII value of G = 71

Practical List : 2

1. Write a program to print the size of various data types in C.

```
#include <stdio.h>
int main()
{
    int integerType;
    float floatType;
    double doubleType;
    char charType;

    // Sizeof operator is used to evaluate the size of a variable
    printf("Size of int: %ld bytes\n",sizeof(integerType));
    printf("Size of float: %ld bytes\n",sizeof(floatType));
    printf("Size of double: %ld bytes\n",sizeof(doubleType));
    printf("Size of char: %ld byte\n",sizeof(charType));

    return 0;
}
```

2. Write a program to check whether the given number is positive or negative.

```
#include <stdio.h>
int main()
{
    double number;

    printf("Enter a number: ");
    scanf("%lf", &number);

    if (number <= 0.0)
    {
        if (number == 0.0)
            printf("You entered 0.");
        else
            printf("You entered a negative number.");
    }
    else
        printf("You entered a positive number.");
    return 0;
}
```

3. Write a program to find out the solution of the quadratic equation of the form $ax^2 + bx + c = 0$.

If discriminant > 0,	$\text{root1} = \frac{-b + \sqrt{(b^2 - 4ac)}}{2a}$ $\text{root2} = \frac{-b - \sqrt{(b^2 - 4ac)}}{2a}$
If discriminant = 0,	$\text{root1} = \text{root2} = \frac{-b}{2a}$
If discriminant < 0,	$\text{root1} = \frac{-b}{2a} + i \frac{\sqrt{-(b^2 - 4ac)}}{2a}$ $\text{root2} = \frac{-b}{2a} - i \frac{\sqrt{-(b^2 - 4ac)}}{2a}$

```
#include <stdio.h>
#include <math.h>
int main()
{
    double a, b, c, discriminant, root1, root2, realPart, imaginaryPart;
    printf("Enter coefficients a, b and c: ");
    scanf("%lf %lf %lf",&a, &b, &c);
    discriminant = b*b-4*a*c;
    // condition for real and different roots
    if (discriminant > 0)
    {
        // sqrt() function returns square root
        root1 = (-b+sqrt(discriminant))/(2*a);
        root2 = (-b-sqrt(discriminant))/(2*a);
        printf("root1 = %.2lf and root2 = %.2lf",root1 , root2);
    }
    //condition for real and equal roots
    else if (discriminant == 0)
    {
        root1 = root2 = -b/(2*a);

        printf("root1 = root2 = %.2lf;", root1);
    }
}
```

```

// if roots are not real
else
{
    realPart = -b/(2*a);
    imaginaryPart = sqrt(-discriminant)/(2*a);
    printf("root1 = %.2lf+%.2lfi and root2 = %.2f-%.2fi", realPart, imaginaryPart, realPart,
imaginaryPart);
}
return 0;
}

```

Output:

Enter coefficients a, b and c: 2.3

4

5.6

Roots are: -0.87+1.30i and -0.87-1.30i

4. Write a program to use specified functions:

ctype.h library functions: isalpha(), isdigit(), islower(), isupper(), tolower(), toupper().

```

#include<stdio.h>
#include<ctype.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c", &ch);

    if(isalnum(ch))
        printf("%c is an alphanumeric character.\n", ch);
    else
        printf("%c is not an alphanumeric character.\n", ch);

    return 0;
}

#include<stdio.h>
#include<ctype.h>

int main()
{
    char ch;

    printf("Enter a character: ");

```

```

scanf("%c",&ch);

if(isalpha(ch))
    printf("%c is an alphabet.\n",ch);
else
    printf("%c is not an alphabet.\n",ch);

return 0;
}

#include<stdio.h>
#include<ctype.h>

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c",&ch);

    if(isdigit(ch))
        printf("%c is a digit.\n",ch);
    else
        printf("%c is not a digit.\n",ch);

    return 0;
}

int main()
{
    char ch;

    printf("Enter a character: ");
    scanf("%c",&ch);

    if(isupper(ch))
        printf("%c is an uppercase character.\n",ch);
    else if(islower(ch))
        printf("%c is an lowercase character.\n",ch);
    else if(isspace(ch))
        printf("%c is space.\n",ch);
    else
        printf("%c is none from uppercase, lowercase and space.\n",ch);

    return 0;
}

#include<stdio.h>
#include<ctype.h>

int main()
{
    char ch;

```

```
printf("Enter a character: ");
scanf("%c", &ch);

if(isprint(ch))
    printf("%c is a printable character.\n", ch);
else
    printf("%c is not a printable character.\n", ch);

return 0;
}
```

Practical List : 3

1. Write a program to find the smallest number out of the given three numbers using nested if...else.

```
#include <stdio.h>

int main()
{
    double n1, n2, n3;
    printf("Enter three numbers: ");
    scanf("%lf %lf %lf", &n1, &n2, &n3);
    if (n1>=n2)
    {
        if(n1>=n3)
            printf("%.2lf is the largest number.", n1);
        else
            printf("%.2lf is the largest number.", n3);
    }
    else
    {
        if(n2>=n3)
            printf("%.2lf is the largest number.", n2);
        else
            printf("%.2lf is the largest number.", n3);
    }
    return 0;
}
```

```
}
```

Output:

Enter three numbers: -4.5

3.9

5.6

5.60 is the largest number.

2. Write a program to design calculator with basic operations using switch case statement.

```
// Performs addition, subtraction, multiplication or division depending the input from user
```

```
# include <stdio.h>
```

```
int main() {
```

```
    char operator;
```

```
    double firstNumber,secondNumber;
```

```
    printf("Enter an operator (+, -, *,): ");
```

```
    scanf("%c", &operator);
```

```
    printf("Enter two operands: ");
```

```
    scanf("%lf %lf",&firstNumber, &secondNumber);
```

```
    switch(operator)
```

```
    {
```

```
        case '+':
```

```
            printf("%.1lf + %.1lf = %.1lf",firstNumber, secondNumber, firstNumber + secondNumber);
```

```
            break;
```

```
        case '-':
```

```
            printf("%.1lf - %.1lf = %.1lf",firstNumber, secondNumber, firstNumber - secondNumber);
```

```
            break;
```

```
        case '*':
```

```
            printf("%.1lf * %.1lf = %.1lf",firstNumber, secondNumber, firstNumber * secondNumber);
```

```
            break;
```

```
        case '/':
```

```

        printf("%.1lf / %.1lf = %.1lf",firstNumber, secondNumber, firstNumber /
secondNumber);

        break;

// operator doesn't match any case constant (+, -, *, /)
default:

        printf("Error! operator is not correct");

    }

    return 0;

}

```

Output:

Enter an operator (+, -, *, /): *

Enter two operands: 1.5

4.5

1.5 * 4.5 = 6.8

3. Write a program to print 1 to N numbers using if and goto statement.

```

#include <stdio.h>

int main()
{
    int counter=1;
    int n;

    //enter the value of n (range)
    printf("Enter the value of n: ");
    scanf("%d",&n);

    //define label
    START:
    printf("%d ",counter);
    counter++; //increment counter

    //condition & goto statement
    if(counter<=n)
        goto START;

    return 0;

}

```

Output:

```
Enter the value of n: 10
1 2 3 4 5 6 7 8 9 10
```

4. Write a program to print table of given number using if and goto statement.

```
#include<stdio.h>

int main()
{
    //declare variable
    int count,t,r;
    //Read value of t
    printf("Enter number: ");
    scanf("%d",&t);
    count=1;
    start:
    if(count<=10)
    {
        //Formula of table
        r=t*count;
        printf("%d*%d=%d\n",t,count,r);
        count++;
        goto start;
    }
    return 0;
}
```

Output

```
Enter number: 10
10*1=10
10*2=20
10*3=30
10*4=40
10*5=50
10*6=60
10*7=70
10*8=80
10*9=90
```

$$10*10=100$$

Practical List : 4

1) Write a program to print 1 to N numbers and sum of them using while, do...while and for loop

```
#include <stdio.h>

int main()
{
    int n, i, sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d",&n);
    for(i=1; i <= n; ++i)
    {
        sum += i; // sum = sum+i;
    }
    printf("Sum = %d",sum);
    return 0;
}
```

2. Write a program to check whether the given number is prime or not using for loop.

```
#include <stdio.h>

int main()
{
    int n, i, flag = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    for(i = 2; i <= n/2; ++i)
    {
        // condition for nonprime number
        if(n%i == 0)
        {
            flag = 1;
        }
    }
}
```

```

        break;
    }
}
if (n == 1)
{
    printf("1 is neither a prime nor a composite number.");
}
else
{
    if (flag == 0)
        printf("%d is a prime number.", n);
    else
        printf("%d is not a prime number.", n);
}
return 0;
}

```

Output

Enter a positive integer: 29
 29 is a prime number.

3. Write a program to reverse a number using while loop.

```

#include <stdio.h>
int main()
{
    int n, reversedNumber = 0, remainder;
    printf("Enter an integer: ");
    scanf("%d", &n);
    while(n != 0)
    {
        remainder = n%10;
        reversedNumber = reversedNumber*10 + remainder;
        n /= 10;
    }
}

```

```

    }

    printf("Reversed Number = %d", reversedNumber);

    return 0;

}

```

Output

Enter an integer: 2345

Reversed Number = 5432

4. Write a program to print the numbers of fibonacci series up to N numbers using do...while loop.

```

#include<stdio.h>
#include<conio.h>
void main()
{
    int n,f,f1=-1,f2=1;
    clrscr();
    printf(" Enter The Number Of Terms:");
    scanf("%d",&n);
    printf(" The Fibonacci Series is:");
    do
    {
        f=f1+f2;
        f1=f2;
        f2=f;
        printf(" \n %d",f);
        n--;
    }while(n>0);
    getch();
}

```

Practical-list-5

1. Write a program to find the sum of all elements of the array.

```

#include <stdio.h>
#define MAX_SIZE 100
int main()
{
    int arr[MAX_SIZE];
    int i, n, sum=0;
    /* Input size of the array */
    printf("Enter size of the array: ");
    scanf("%d", &n);

    /* Input elements in array */
    printf("Enter %d elements in the array: ", n);
}

```

```

for(i=0; i<n; i++)
{
    scanf("%d", &arr[i]);
}

/*
 * Add each array element to sum
 */
for(i=0; i<n; i++)
{
    sum = sum + arr[i];
}
printf("Sum of all elements of array = %d", sum);

return 0;
}

```

2. Write a program to sort N numbers and display on the screen.

```

#include <stdio.h>
void main ()
{
    int number[30];
    int i, j, a, n;
    printf("Enter the value of N\n");
    scanf("%d", &n);
    printf("Enter the numbers \n");
    for (i = 0; i < n; ++i)
        scanf("%d", &number[i]);
    for (i = 0; i < n; ++i)
    {
        for (j = i + 1; j < n; ++j)
        {
            if (number[i] > number[j])
            {
                a = number[i];
                number[i] = number[j];
                number[j] = a;
            }
        }
    }
}

```

```

    }
}
printf("The numbers arranged in descending order are given below\n");
for (i = 0; i < n; ++i)
{
    printf("%d\n", number[i]);
}
}

```

3. Write a program to find out the largest number in an array of n integers.

```

#include <stdio.h>
int main()
{
    int i, n;
    float arr[100];
    printf("Enter total number of elements(1 to 100): ");
    scanf("%d", &n);
    printf("\n");
    // Stores number entered by the user
    for(i = 0; i < n; ++i)
    {
        printf("Enter Number %d: ", i+1);
        scanf("%f", &arr[i]);
    }
    // Loop to store largest number to arr[0]
    for(i = 1; i < n; ++i)
    {
        // Change < to > if you want to find the smallest element
        if(arr[0] < arr[i])
            arr[0] = arr[i];
    }
    printf("Largest element = %.2f", arr[0]);
    return 0;
}

```

Output

Enter total number of elements(1 to 100): 8

Enter Number 1: 23.4

Enter Number 2: -34.5

Enter Number 3: 50

Enter Number 4: 33.5

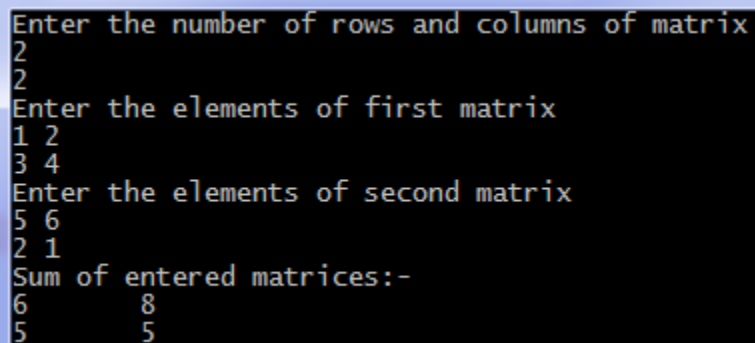
Enter Number 5: 55.5

Enter Number 6: 43.7
Enter Number 7: 5.7
Enter Number 8: -66.5

4. Write a program for addition of two 3x3 matrix.

```
#include <stdio.h>
int main()
{
    int m, n, c, d, first[10][10], second[10][10], sum[10][10];
    printf("Enter the number of rows and columns of matrix\n");
    scanf("%d%d", &m, &n);
    printf("Enter the elements of first matrix\n");
    for (c = 0; c < m; c++)
        for (d = 0; d < n; d++)
            scanf("%d", &first[c][d]);
    printf("Enter the elements of second matrix\n");
    for (c = 0; c < m; c++)
        for (d = 0; d < n; d++)
            scanf("%d", &second[c][d]);
    printf("Sum of entered matrices:-\n");
    for (c = 0; c < m; c++) {
        for (d = 0; d < n; d++) {
            sum[c][d] = first[c][d] + second[c][d];
            printf("%d\t", sum[c][d]);
        }
        printf("\n");
    }
}
```

Output



```
Enter the number of rows and columns of matrix
2
2
Enter the elements of first matrix
1 2
3 4
Enter the elements of second matrix
5 6
2 1
Sum of entered matrices:-
6      8
5      5
```

5. Write a program to count number of vowels in a given string.

```
#include <stdio.h>
int main()
```

```

{
    char line[150];
    int i, vowels;
    vowels = 0;
    printf("Enter a line of string: ");
    scanf("%[^\n]", line);
    for(i=0; line[i]!='\0'; ++i)
    {
        if((line[i]=='a' || line[i]=='e' || line[i]=='i' ||
            line[i]=='o' || line[i]=='u' || line[i]=='A' ||
            line[i]=='E' || line[i]=='I' || line[i]=='O' ||
            line[i]=='U'))
        {
            ++vowels;
        }
    }
    printf("Vowels: %d",vowels);
    return 0;
}

```

6. Write a program to perform following string handling functions. (With use of <string.h>) (strcpy() , strlen(),strcmp(),strcat())

Strcpy:

```

#include <stdio.h>
#include <string.h>
int main()
{
    char str1[10]= "awesome";
    char str2[10];
    char str3[10];

    strcpy(str2, str1);
    strcpy(str3, "well");
    puts(str2);
    puts(str3);

    return 0;
}

```

Strlen:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char c[100];
    printf("Enter string: ");
    gets(c);
    printf("Length of string a = %d \n",strlen(c));
    return 0;
}
```

Strcmp:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str1[] = "nehal", str2[] = "sNeh", str3[] = "nehal";
    int result;
    result = strcmp(str1, str2);
    printf("strcmp(str1, str2) = %d\n", result);

    result = strcmp(str1, str3);
    printf("strcmp(str1, str3) = %d\n", result);

    return 0;
}
```

Strcat:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str1[] = "Charusat", str2[] = "Department of I.T.";
    strcat(str1, str2);
}
```

```
puts(str1);
printf("\n");
puts(str2);
return 0;
}
```

7. Write a program to accept two strings and concatenate them without string handling functions / in-built library functions.

//Accept two strings and concatenate them.

```
#include <stdio.h>
int main()
{
    char s1[100], s2[100], i, j;

    printf("Enter first string: ");
    scanf("%s", s1);

    printf("Enter second string: ");
    scanf("%s", s2);

    for(i = 0; s1[i] != '\0'; ++i);

    for(j = 0; s2[j] != '\0'; ++j, ++i)
    {
        s1[i] = s2[j];
    }

    s1[i] = '\0';
    printf("After concatenation: %s", s1);

    return 0;
}
```

8. Write a program for multiplication of two 3x3 matrix. (Hint: $A_{ij} * B_{jk} = C_{ik}$).

```
#include<stdio.h>
int main()
{
    int a[10][10], b[10][10], c[10][10], i, j, k,sum=0;
    printf("\nEnter First Matrix : n");
```

```

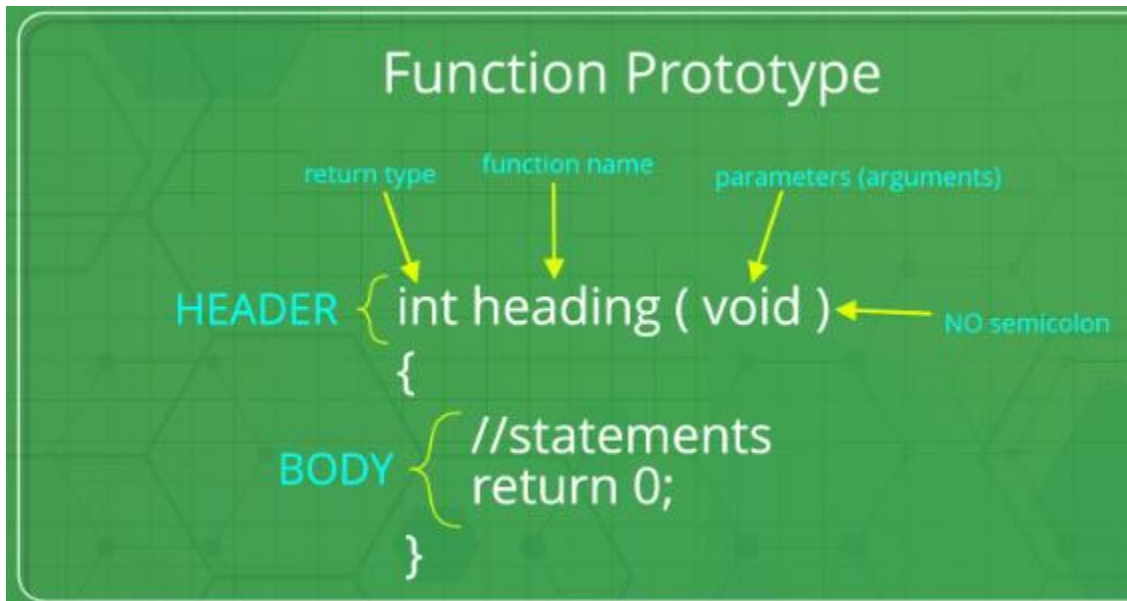
for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        scanf("%d", &a[i][j]);
    }
}
printf("\nEnter Second Matrix:");
for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        scanf("%d", &b[i][j]);
    }
}
for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        sum = 0;
        for (k = 0; k < 3; k++)
        {
            sum = sum + a[i][k]*b[k][j];
        }
        c[i][j] = sum;
        printf(" %d ", c[i][j]);
    }
    printf("\n");
}
return (0);
}

```

Practical-list-6

1. Write a program to multiply two numbers using function with the help of following criteria:

- a. Function with no argument, no return value
- b. Function with an argument, but no return value
- c. Function with no argument, but a return value
- d. Function with an argument and a return value



```
Function declaration : void function();  
Function call : function();  
Function definition :  
    void function()  
    {  
        statements;  
    }
```

2. Write a program to calculate nCr using function. (Hint: $nCr = n! / ((r!) (n - r)!)$)

```
#include <stdio.h>  
int fact(int z);  
void main()  
{  
    int n, r, ncr;
```

```

printf("\n Enter the value for N and R \n");
scanf("%d%d", &n, &r);
ncr = fact(n) / (fact(r) * fact(n - r));
printf("\n The value of ncr is: %d", ncr);
}

int fact(int z)
{
    int f = 1, i;
    if (z == 0)
    {
        return(f);
    }
    else
    {
        for (i = 1; i <= z; i++)
        {
            f = f * i;
        }
    }
    return(f);
}

```

3. Write a program to generate Fibonacci numbers using recursive function.

```

#include<stdio.h>
int Fibonacci(int);
int main()
{
    int n, i = 0, c;
    scanf("%d",&n);
    printf("Fibonacci series\n");
    for ( c = 1 ; c <= n ; c++ )
    {
        printf("%d\n", Fibonacci(i));
    }
}

```

```

        i++;
    }
    return 0;
}
int Fibonacci(int n)
{
    if ( n == 0 )
        return 0;
    else if ( n == 1 )
        return 1;
    else
        return ( Fibonacci(n-1) + Fibonacci(n-2) );
}

```

4. Define a structure data type called time_struct containing three members integer hour, integer minute and integer second. Develop a program that would assign values to the individual members & display the time in the following from: 16:40:51

```

#include <stdio.h>
struct time_struct
{
    int hour;
    int minute;
    int second;
};
void main()
{
    struct time_struct time;
    clrscr();
    time.hour=16;
    time.minute=40;
    time.second=51;
    printf("%d:%d:%d",time.hour,time.minute,time.second);
    getch();
}

```

```
}
```

OUTPUT

16:40:51

6. Write a program to create a structure of an employee with full name, city and salary. Display it for n employees. (Hint: Use the concept of nested structure).

```
#include<stdio.h>

struct Address
{
    char HouseNo[25];
    char City[25];
    char PinCode[25];
};

struct Employee
{
    int Id;
    char Name[25];
    float Salary;
    struct Address Add;
};

void main()
{
    int i;
    struct Employee E;
    printf("\n\tEnter Employee Id : ");
    scanf("%d",&E.Id);
    printf("\n\tEnter Employee Name : ");
    scanf("%s",&E.Name);
    printf("\n\tEnter Employee Salary : ");
    scanf("%f",&E.Salary);
    printf("\n\tEnter Employee House No : ");
    scanf("%s",&E.Add.HouseNo);
    printf("\n\tEnter Employee City : ");
    scanf("%s",&E.Add.City);
```

```

printf("\n\tEnter Employee House No : ");
scanf("%s",&E.Add.PinCode);
printf("\nDetails of Employees");
printf("\n\tEmployee Id : %d",E.Id);
printf("\n\tEmployee Name : %s",E.Name);
printf("\n\tEmployee Salary : %f",E.Salary);
printf("\n\tEmployee House No : %s",E.Add.HouseNo);
printf("\n\tEmployee City : %s",E.Add.City);
printf("\n\tEmployee House No : %s",E.Add.PinCode);
}

```

Output :

```

Enter Employee Id : 101
Enter Employee Name : Suresh
Enter Employee Salary : 45000
Enter Employee House No : 4598/D
Enter Employee City : Delhi
Enter Employee Pin Code : 110056

```

Details of Employees

```

Employee Id : 101
Employee Name : Suresh
Employee Salary : 45000
Employee House No : 4598/D
Employee City : Delhi
Employee Pin Code : 110056

```

7. Write a program to addition of two numbers using pointer.

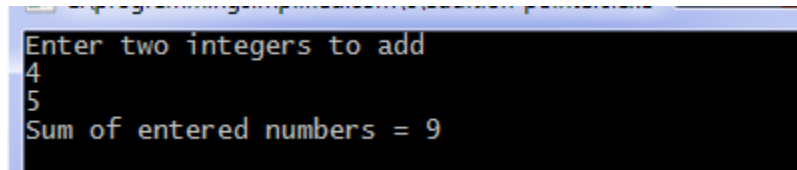
```

#include <stdio.h>
int main()
{
    int first, second, *p, *q, sum;
    printf("Enter two integers to add\n");
    scanf("%d%d", &first, &second);

```

```
p = &first;  
q = &second;  
sum = *p + *q;  
printf("Sum of the numbers = %d\n", sum);  
return 0;  
}
```

Output

A screenshot of a terminal window with a black background and white text. The text shows the program's execution: a prompt to enter two integers, the user inputting 4 and 5, and the program outputting the sum 9.

```
Enter two integers to add  
4  
5  
Sum of entered numbers = 9
```